Solution

Laplace Inverse Transform of $\frac{x-6}{x\left(x^2+3x+5\right)}$: $-\frac{6}{5}\mathrm{H}(t)+\frac{6}{5}e^{-\frac{3t}{2}}\cos\left(\frac{\sqrt{11}\,t}{2}\right)+\frac{28}{5\sqrt{11}}e^{-\frac{3t}{2}}\sin\left(\frac{\sqrt{11}\,t}{2}\right)$

Steps

$L^{-1}\left\{\frac{x-6}{x\left(x^2+3x+5\right)}\right\}$

Take the partial fraction of $\frac{x-6}{x\left(x^2+3x+5\right)}$: $-\frac{6}{5x}+\frac{6x+23}{5\left(x^2+3x+5\right)}$ *Show Steps*

$=L^{-1}\left\{-\frac{6}{5x}+\frac{6x+23}{5\left(x^2+3x+5\right)}\right\}$

Expand $\frac{6x+23}{5\left(x^2+3x+5\right)}$: $\frac{6}{5}\cdot\frac{x+\frac{3}{2}}{\left(x+\frac{3}{2}\right)^2+\frac{11}{4}}+\frac{14}{5}\cdot\frac{1}{\left(x+\frac{3}{2}\right)^2+\frac{11}{4}}$ *Show Steps*

$=L^{-1}\left\{-\frac{6}{5x}+\frac{6}{5}\cdot\frac{x+\frac{3}{2}}{\left(x+\frac{3}{2}\right)^2+\frac{11}{4}}+\frac{14}{5}\cdot\frac{1}{\left(x+\frac{3}{2}\right)^2+\frac{11}{4}}\right\}$

Use the linearity property of Inverse Laplace Transform:
For functions $f(s), g(s)$ and constants a, b: $L^{-1}\{a\cdot f(s)+b\cdot g(s)\}=a\cdot L^{-1}\{f(s)\}+b\cdot L^{-1}\{g(s)\}$

$=-L^{-1}\left\{\frac{6}{5x}\right\}+\frac{6}{5}L^{-1}\left\{\frac{x+\frac{3}{2}}{\left(x+\frac{3}{2}\right)^2+\frac{11}{4}}\right\}+\frac{14}{5}L^{-1}\left\{\frac{1}{\left(x+\frac{3}{2}\right)^2+\frac{11}{4}}\right\}$

$L^{-1}\left\{\frac{6}{5x}\right\}$: $\frac{6}{5}\mathrm{H}(t)$ *Show Steps*

$L^{-1}\left\{\frac{x+\frac{3}{2}}{\left(x+\frac{3}{2}\right)^2+\frac{11}{4}}\right\}$: $e^{-\frac{3t}{2}}\cos\left(\frac{\sqrt{11}\,t}{2}\right)$ *Show Steps*

$L^{-1}\left\{\frac{1}{\left(x+\frac{3}{2}\right)^2+\frac{11}{4}}\right\}$: $e^{-\frac{3t}{2}}\frac{2}{\sqrt{11}}\sin\left(\frac{\sqrt{11}\,t}{2}\right)$ *Show Steps*

$=-\frac{6}{5}\mathrm{H}(t)+\frac{6}{5}e^{-\frac{3t}{2}}\cos\left(\frac{\sqrt{11}\,t}{2}\right)+\frac{14}{5}e^{-\frac{3t}{2}}\frac{2}{\sqrt{11}}\sin\left(\frac{\sqrt{11}\,t}{2}\right)$

Refine $-\frac{6}{5}\mathrm{H}(t)+\frac{6}{5}e^{-\frac{3t}{2}}\cos\left(\frac{\sqrt{11}\,t}{2}\right)+\frac{14}{5}e^{-\frac{3t}{2}}\frac{2}{\sqrt{11}}\sin\left(\frac{\sqrt{11}\,t}{2}\right)$: $-\frac{6}{5}\mathrm{H}(t)+\frac{6}{5}e^{-\frac{3t}{2}}\cos\left(\frac{\sqrt{11}\,t}{2}\right)$ Show Steps

$$=-\frac{6}{5}\mathrm{H}(t)+\frac{6}{5}e^{-\frac{3t}{2}}\cos\left(\frac{\sqrt{11}\,t}{2}\right)+\frac{28}{5\sqrt{11}}e^{-\frac{3t}{2}}\sin\left(\frac{\sqrt{11}\,t}{2}\right)$$